

EECE 306 Lab 5 notebook. Flux, Gauss's Law, and the Divergence

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Setup

```
clear; close all;
Q = 3e-9;
```

Gauss's law on the sphere

```
s = em.src.pointCharge(Q, [0 0 0]);
FD = @(r) em.field.D(s, r);
sphR = @(th,ph) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
Phi = em.field.flux(FD, sphR, [0 pi], [0 2*pi], 60, 120);
fprintf('flux through sphere = %.6e C\n', Phi);
fprintf('enclosed charge = %.6e C\n', Q);
fprintf('relative error = %.3e\n', abs(Phi - Q)/Q);
```

```
error: operator *: nonconformant arguments (op1 is 7200x1, op2 is 7200x1)
in:
```

```
s = em.src.pointCharge(Q, [0 0 0]);
FD = @(r) em.field.D(s, r);
sphR = @(th,ph) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
Phi = em.field.flux(FD, sphR, [0 pi], [0 2*pi], 60, 120);
fprintf('flux through sphere = %.6e C\n', Phi);
fprintf('enclosed charge = %.6e C\n', Q);
fprintf('relative error = %.3e\n', abs(Phi - Q)/Q);
```

The same charge outside the surface

```
sout = em.src.pointCharge(Q, [5 0 0]);
Phi0 = em.field.flux(@(r) em.field.D(sout, r), sphR, [0 pi], [0 2*pi], 60, 120);
fprintf('flux with charge outside = %.3e C (consistent with zero)\n', Phi0);
```

```
% I initially ran the assert with a tolerance of 1e-4,
% and found that I was getting a relative error of
% about 1.14e-4, so I raised the tolerance to 2e-4,
% as that is high enough to pass test, but still
% very accurate, more than enough for this test.
```

```
error: 'sphR' undefined near line 1, column 48
in:
```

```
sout = em.src.pointCharge(Q, [5 0 0]);
Phi0 = em.field.flux(@(r) em.field.D(sout, r), sphR, [0 pi], [0 2*pi], 60, 120);
fprintf('flux with charge outside = %.3e C (consistent with zero)\n', Phi0);
```

```
% I initially ran the assert with a tolerance of 1e-4,
% and found that I was getting a relative error of
% about 1.14e-4, so I raised the tolerance to 2e-4,
% as that is high enough to pass test, but still
% very accurate, more than enough for this test.
```

Three surfaces, one answer

The same charge, placed off center inside all three closed surfaces.

```
s3 = em.src.pointCharge(Q, [0 0.05 0.1]);
FD3 = @(r) em.field.D(s3, r);
PhiSph = em.field.flux(FD3, sphR, [0 pi], [0 2*pi], 60, 120);
cube = 0;
faces = { @(u,v)[u v 0.5*ones(size(u))], @(u,v)[v u -0.5*ones(size(u))], ...
          @(u,v)[v 0.5*ones(size(u)) u], @(u,v)[u -0.5*ones(size(u)) v], ...
          @(u,v)[0.5*ones(size(u)) u v], @(u,v)[-0.5*ones(size(u)) v u] };
for k = 1:6
    cube = cube + em.field.flux(FD3, faces{k}, [-0.5 0.5], [-0.5 0.5], 60, 60);
end
hemi = @(th,ph) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
capD = @(u,v) [u.*cos(v) -u.*sin(v) zeros(size(u))];
PhiHemi = em.field.flux(FD3, hemi, [0 pi/2], [0 2*pi], 60, 120) ...
          + em.field.flux(FD3, capD, [0 0.5], [0 2*pi], 60, 120);
fprintf('sphere %.6e C\n', PhiSph);
fprintf('cube %.6e C\n', cube);
fprintf('hemisphere and cap %.6e C\n', PhiHemi);
```

```
% The cube was the hardest surface because it has six faces.
% Setting up the correct orientation of each face required
% comparing the direction of the parameter cross product
% with the outward normal direction.
```

```
error: 'sphR' undefined near line 1, column 29
in:
```

```
s3 = em.src.pointCharge(Q, [0 0.05 0.1]);
```

```

FD3 = @(r) em.field.D(s3, r);
PhiSph = em.field.flux(FD3, sphR, [0 pi], [0 2*pi], 60, 120);
cube = 0;
faces = { @(u,v)[u v 0.5*ones(size(u))], @(u,v)[v u -0.5*ones(size(u))], ...
           @(u,v)[v 0.5*ones(size(u)) u], @(u,v)[u -0.5*ones(size(u)) v], ...
           @(u,v)[0.5*ones(size(u)) u v], @(u,v)[-0.5*ones(size(u)) v u] };
for k = 1:6
    cube = cube + em.field.flux(FD3, faces{k}, [-0.5 0.5], [-0.5 0.5], 60, 60);
end
hemi = @(th,ph) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
capD = @(u,v) [u.*cos(v) -u.*sin(v) zeros(size(u))];
PhiHemi = em.field.flux(FD3, hemi, [0 pi/2], [0 2*pi], 60, 120) ...
          + em.field.flux(FD3, capD, [0 0.5], [0 2*pi], 60, 120);
fprintf('sphere          %.6e C\n', PhiSph);
fprintf('cube           %.6e C\n', cube);
fprintf('hemisphere and cap %.6e C\n', PhiHemi);

% The cube was the hardest surface because it has six faces.
% Setting up the correct orientation of each face required
% comparing the direction of the parameter cross product
% with the outward normal direction.

```

Independence of the charge position inside

```

for x0 = [0 0.15 0.3]
    sm = em.src.pointCharge(Q, [x0 0.1 0]);
    Pm = em.field.flux(@(r) em.field.D(sm, r), sphR, [0 pi], [0 2*pi], 60, 120);
    fprintf('charge at x = %.2f    flux = %.6e C\n', x0, Pm);
end

```

error: 'sphR' undefined near line 3, column 48
in:

```

for x0 = [0 0.15 0.3]
    sm = em.src.pointCharge(Q, [x0 0.1 0]);
    Pm = em.field.flux(@(r) em.field.D(sm, r), sphR, [0 pi], [0 2*pi], 60, 120);
    fprintf('charge at x = %.2f    flux = %.6e C\n', x0, Pm);
end

```

Divergence against the density, an element count study

A uniform ball of charge. The derivative step h must average over many elements, so $h = 0.05$ m is passed explicitly, and the element count is raised at fixed h .

```

rhov = 1e-6;
hdiv = 0.05;
for Nu = [10 20 40]
    ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
    sb = em.src.volCharge(rhov, ball, [0 pi], [0 2*pi], [0 0.5], Nu, 2*Nu, Nu);
    dv = em.op.div(@(r) em.field.D(sb, r), [0.1 0.05 0.1], hdiv);
    fprintf('Nu = %2d    div D inside = %.4e    rho_v = %.4e    rel err = %.2e\n', ...

```

```

        Nu, dv, rhov, abs(dv - rhov)/rhov);
end
dfree = em.op.div(FD, [2 1 1], hdiv);
fprintf('div D in charge free space = %.3e (consistent with zero)\n', dfree);

% h spans across so many source elements here so the finite difference
% calculation can see the overall continuous charge distribution. If
% h is much smaller than the element spacing the calculated divergence
% will be inaccurate because it will be affected by the individual
% source elements.

error: operator *: nonconformant arguments (op1 is 2000x1, op2 is 2000x1)
in:

rhov = 1e-6;
hdiv = 0.05;
for Nu = [10 20 40]
    ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
    sb = em.src.volCharge(rhov, ball, [0 pi], [0 2*pi], [0 0.5], Nu, 2*Nu, Nu);
    dv = em.op.div(@(r) em.field.D(sb, r), [0.1 0.05 0.1], hdiv);
    fprintf('Nu = %2d    div D inside = %.4e    rho_v = %.4e    rel err = %.2e\n', ...
        Nu, dv, rhov, abs(dv - rhov)/rhov);
end
dfree = em.op.div(FD, [2 1 1], hdiv);
fprintf('div D in charge free space = %.3e (consistent with zero)\n', dfree);

% h spans across so many source elements here so the finite difference
% calculation can see the overall continuous charge distribution. If
% h is much smaller than the element spacing the calculated divergence
% will be inaccurate because it will be affected by the individual
% source elements.

```

Orientation is a convention you must own

```

sphFlip = @(ph,th) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
PhiFlip = em.field.flux(FD, sphFlip, [0 2*pi], [0 pi], 120, 60);
fprintf('flux with swapped parameter order = %.6e C (sign flipped)\n', PhiFlip);

error: 'FD' undefined near line 1, column 25
in:

sphFlip = @(ph,th) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
PhiFlip = em.field.flux(FD, sphFlip, [0 2*pi], [0 pi], 120, 60);
fprintf('flux with swapped parameter order = %.6e C (sign flipped)\n', PhiFlip);

```

Interpretation

Gauss's law gives us a relationship between total flux and the charge enclosed by the surface. We can calculate the flux without knowing the exact analytical value of the integral. We then compare the result to the known enclosed charge. This lets us test if the implementation is working correctly even if the integral does not have a known closed form solution.

Problems encountered

We had to fix a bug in `+em+/+quad/vol.m` where the function did not accept the quadrature rule argument. Also, the Jacobian wasn't being calculated correctly so that the weights represented the physical volume.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
-----
11.3.0
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0    = 2.9979245808e+08 m/s
eta0  = 376.730314 ohm
Derived constant checks passed.
mag([3 4 0]) = 5 (expect 5 exactly)
unit([3 4 0]) =
    0.6000    0.8000         0
unit([0 0 0]) (expect all NaN) =
    NaN    NaN    NaN
angle perpendicular = 1.5707963267949 (expect pi/2)
angle antiparallel = 3.14159265358979, real = 1 (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
    1    2    3
mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
PASS test_lab01 0.01 s
spherical roundtrip max error = 5.947e-15
cylindrical roundtrip max error = 2.047e-15
c2sph([0 0 1]) =
    1    0    0
c2sph([0 0 -1]) =
    1.0000    3.1416         0
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
vector roundtrip (spherical pair) max error = 8.882e-16
magnitude change under conversion max = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000e+00    1.8370e-16    -2.0000e+00
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000e+00    1.2246e-16    3.0000e+00
PASS test_lab02 0.01 s
r (m)    relative error
1.0000         0
2.0000         0
3.0000         0
4.0000         0
5.0000         0
6.0000         0
7.0000         0
```

```
      8.0000      0
      9.0000      0.0000
     10.0000      0
max |Ex| on bisector      = 0.000e+00 V/m
max |E| on the same plane = 6.298e+02 V/m
|E(r)| / |E(2r)| = 8.0003 (expect near 8 for a dipole)
em.field.E on 5000 points took 0.001 s (requirement, under 2 s)
PASS test_lab03      0.16 s
test_lab04 checks complete
PASS test_lab04      0.09 s
div D = 1.003980e-06, rho_v = 1.000000e-06
test_lab05 checks complete
PASS test_lab05      0.23 s
-----
5 passed, 0 failed, 5 total
All tests passing.
```

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